5731.25 Wireless Communication — Trial Exam 9

26. March 2026

Instructions to Students

- Ensure clarity and precision in your answers.
- Attempt all sections. Each of the three sections contributes one-third (1/3) of the total mark.
- All answers must be written in **English**.
- Write your answers in the provided answer sheets using a pen.
- Each page must contain your name and student number.
- Number all submitted pages as *Page X/Y*.
- For the multiple-choice section, indicate your answers using the format: Q1 A, Q2 C, ...
- In the appendix of this document, you will find a logarithm table for reference.

Section 1 — Multiple Choice (20 questions)

Choose the **one best answer** for each question.

1. The unit dBm is a measure of:
   a) power referenced to 1 W
   b) power referenced to 1 mW
   c) voltage referenced to 1 V
   d) path loss

2. Thermal noise density at room temperature is approximately:
   a) -174 dBm/Hz
   b) -120 dBm/Hz
   c) 0 dBm/Hz
   d) $+10$ dBm/Hz

3. If symbol rate R_s doubles, bandwidth B (rule of thumb) tends to:
   a) stay constant
   b) decrease
   c) double
   d) increase quadratically with R_s

4. Increasing modulation order M tends to:
   a) increase bits per symbol but require higher SNR and/or more bandwidth
   b) reduce bits per symbol and improve robustness
   c) eliminate the need for bandwidth
   d) make SNR irrelevant

5. Duplexing defines:
   a) whether transmit and receive can occur simultaneously
   b) how to compute link budget losses
   c) how many taps are used in a channel model
   d) the relationship between T_s and R_s

6. A key challenge for resource allocation (besides interference management) is:
   a) quality of service (QoS) requirements like low latency and fairness
   b) ignoring scalability
   c) infinite wireless spectrum availability
   d) removing all multipath automatically

7. Which propagation mechanism allows a signal to bend around edges?
   a) Reflection
   b) Diffraction
   c) Refraction
   d) Polarization

8. Beamforming mainly provides:
   a) random delay
   b) directional gain
   c) lower bandwidth
   d) no need for antennas

9. In unlicensed bands, interference is typically:
   a) lower due to exclusive access
   b) absent because of regulation
   c) higher due to shared access
   d) fixed and predictable

10. Duplexing in wireless communication refers to:
    a) whether a system can send and receive data at the same time
    b) how noise figure is calculated
    c) how to compute free-space path loss
    d) only how modulation shifts signals to higher frequencies

11. SNR in dB is computed as:
    a) $10\log_{10}(P_s/P_n)$
    b) $20\log_{10}(P_s/P_n)$
    c) $10\log_{10}(P_n/P_s)$

d) P_s/P_n

12. Which modulation carries the most bits/symbol?
    a) BPSK
    b) QPSK
    c) 16-QAM
    d) all equal

13. Free-space path loss increases when:
    a) frequency decreases
    b) distance decreases
    c) frequency or distance increases
    d) transmit power increases

14. A narrowband cellular IoT technology is:
    a) NB-IoT
    b) Wi-Fi
    c) GPS
    d) FM radio

15. Frequency reuse in cellular systems is used to:
    a) eliminate all interference
    b) increase spatial reuse of spectrum
    c) remove fading
    d) replace coding

16. Which item most directly improves link budget?
    a) larger packet header
    b) higher antenna gain
    c) longer cable
    d) lower sampling rate

17. A link is feasible when:
    a) $P_t > 0$ dBm
    b) received SINR exceeds required threshold
    c) carrier is below 1 GHz
    d) antennas have equal gain

18. Which statement best explains why media access control is hard in wireless networks?
    a) transmitters can always detect collisions on the shared medium
    b) transmitters often cannot detect collisions
    c) interference is eliminated by increasing bandwidth
    d) noise is never present in wireless channels

19. Which is an example of licensed spectrum usage?
    a) NB-IoT
    b) Wi-Fi at 2.4 GHz
    c) Bluetooth LE
    d) Zigbee

20. If transmit power and path losses are fixed, increasing bandwidth can lead to:
    a) higher noise power, reducing SNR (so capacity does not necessarily increase)
    b) lower noise power, so SNR always increases
    c) no change in noise power, so SNR stays constant
    d) capacity becomes independent of bandwidth

Section 2 — Short Questions (6 questions)

Answer briefly but clearly.

1. Define symbol rate R_s and relate it to T_s.
2. Explain the effect of increasing R_s on the required bandwidth B.
3. Spectral efficiency: what does increasing M and/or R_s tend to require in terms of SNR and/or bandwidth?
4. Define duplexing.
5. Resource allocation: what is the purpose of allocating resources among multiple users?
6. ISI: give the main reason symbols can overlap at the receiver.

Section 3 — Long / Applied Questions

Show all calculations and reasoning.

Question 1 — Link Budget and Feasibility

A transmitter operates at $f = 880$ MHz with:

- transmit power $P_t = 16$ dBm,
- transmit antenna gain $G_t = 2$ dBi,
- receive antenna gain $G_r = 2$ dBi,
- distance $d = 1.0$ km.

Assume:

$$N = -115 \text{ dBm}, \qquad \gamma_{\min} = 5 \text{ dB}$$

a) Compute the free-space path loss at 880 MHz.

b) Compute the received power P_r.

c) Compute the minimum required received power for feasibility.

d) Is the link feasible?

Question 2 — Indoor Path Loss

An indoor link has:

- distance-based path loss of 73 dB,
- one concrete wall causing 10 dB loss,
- one concrete wall causing 8 dB loss,
- one wooden door causing 2 dB loss,
- one glass window causing 3 dB loss.

Transmitter and receiver antenna gains are 0 dBi, and $P_t = 17$ dBm.

a) Compute total path loss.

b) Compute received power.

c) If receiver sensitivity is -78 dBm, does the link work?

d) Suggest two simple ways to improve this link without changing bandwidth.

Question 3 — Frequency Reuse

Assume a cellular layout with reuse factor $N = 3$.

a) What fraction of total spectrum is available in each cell?

b) Compare this with a system using $N = 5$.

c) Explain how changing N impacts:

   i) co-channel interference,

   ii) capacity per cell.

Question 4 — Multi-hop Routing

Only the following links are feasible:

$$1 \to 2, \quad 1 \to 3, \quad 2 \to 3, \quad 2 \to 4, \quad 3 \to 4, \quad 1 \to 4$$

Transmission energy costs are:

$$E_{12} = 2.9, \quad E_{13} = 2.2, \quad E_{23} = 1.5, \quad E_{24} = 2.6, \quad E_{34} = 2.1, \quad E_{14} = 6.1$$

where node 1 is source and node 4 is gateway.

a) List all feasible routes from node 1 to node 4.

b) Compute total energy for each route.

c) Identify the minimum-energy route.

d) Give one advantage and one disadvantage of multi-hop communication.

Question 5 — Modulation and Throughput

A wireless channel has bandwidth $B = 130$ kHz.

Figure 1 shows throughput (bits/s/Hz) versus SNR for several candidate modulation/coding schemes.

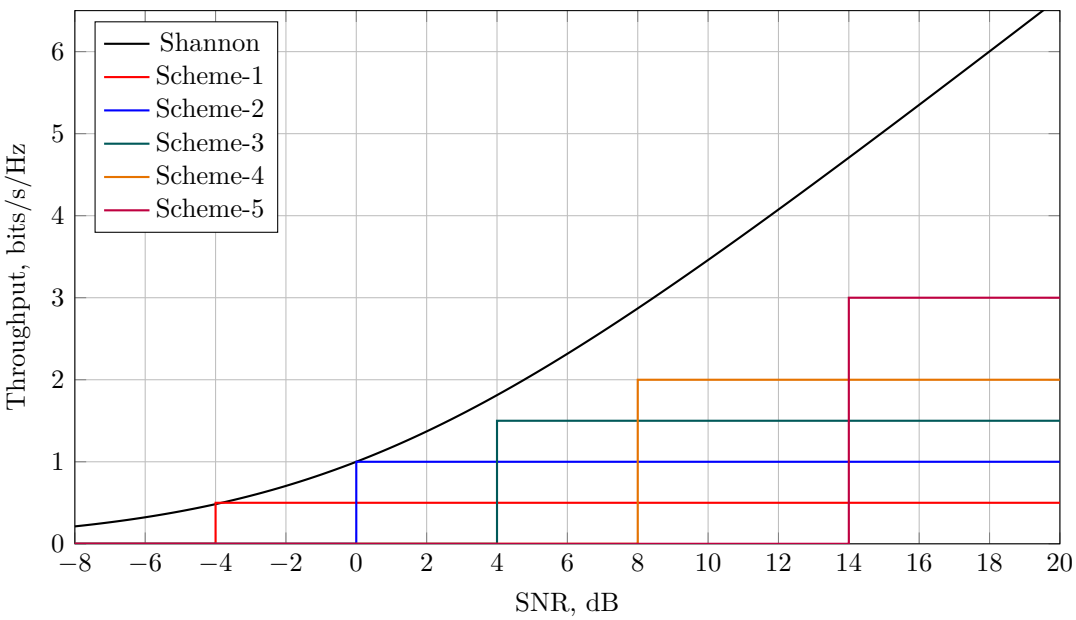


Figure 1: Throughput versus SNR for candidate schemes and Shannon bound.

a) For SNR = 16 dB, compute Shannon capacity in bit/s.

b) From Figure 1, choose the highest usable scheme at 16 dB and compute data rate in bit/s.

c) At 18 dB the system uses scheme-5. How much SNR drop (in dB) is possible before switching down to scheme-4?

Question 6 — Coverage Planning

The terrain sketch on the last page shows two towns separated by a hill.

a) Propose a single base-station location and explain your choice.

b) Propose a two-site solution (base station + relay) and explain the benefit.

c) Compare the two solutions in terms of coverage, cost, and reliability.

d) Which propagation effects are most important in this scenario?

End of exam

1 Supplemental material

1.1 Logarithm Properties

- **Product Rule:**

$$\log_b(xy) = \log_b(x) + \log_b(y)$$

- **Quotient Rule:**

$$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$$

- **Power Rule:**

$$\log_b(x^n) = n\log_b(x)$$

- **Base Change Formula:**

$$\log_b(x) = \frac{\log_c(x)}{\log_c(b)}$$

1.2 Common Constants

- **Speed of Light:**

$$c \approx 3.0 \times 10^8 \text{ m/s}$$

- **Thermal Noise Density (room temperature):**

$$P_n(1 \text{ Hz}) \approx -174 \text{ dBm/Hz}$$

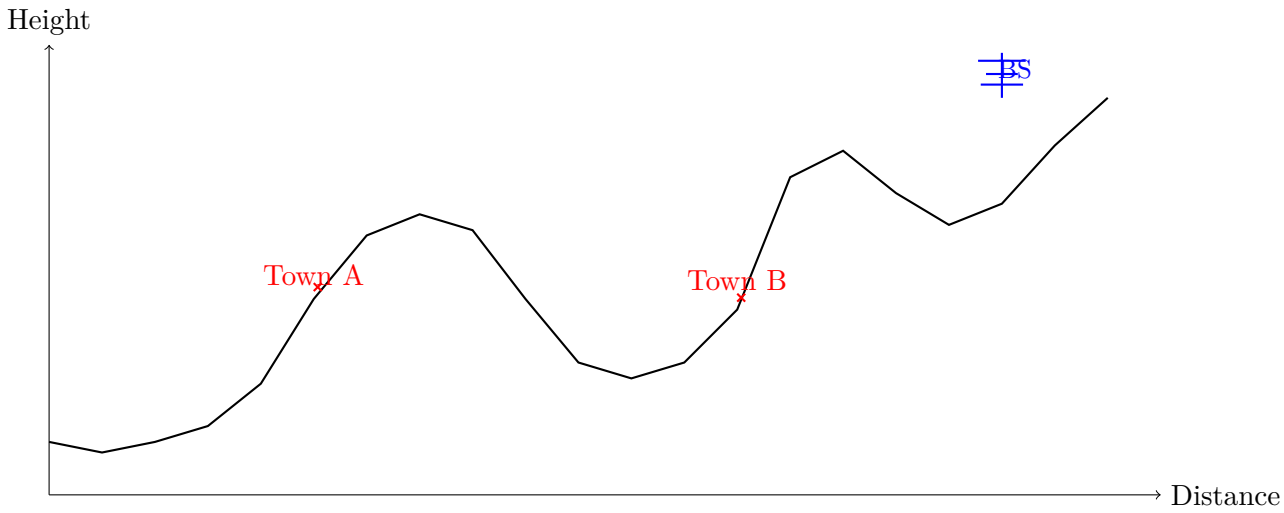
1.3 Useful Log Values

Value	$\log_{10}$(Value)	$\log_2$(Value)	Scientific Notation
2	0.301	1.000	2.0×10^0
3	0.477	1.585	3.0×10^0
4	0.602	2.000	4.0×10^0
5	0.699	2.322	5.0×10^0
6	0.778	2.585	6.0×10^0
7	0.845	2.807	7.0×10^0
8	0.903	3.000	8.0×10^0
9	0.954	3.170	9.0×10^0
10	1.000	3.322	1.0×10^1

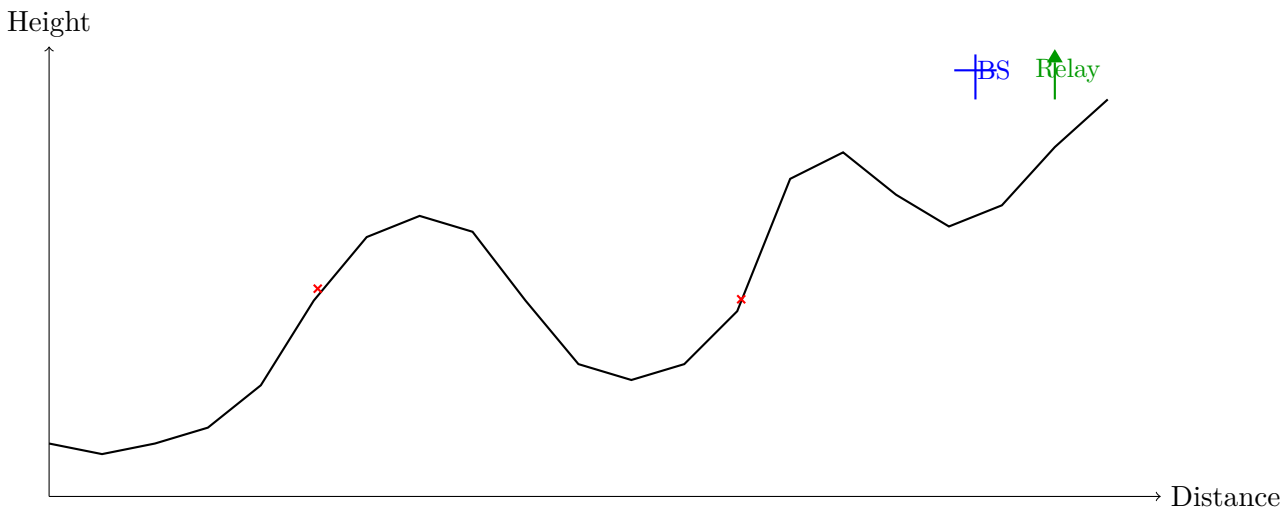
Answer Page — Coverage Planning

Name: _____ **Page** [ ] **of** [ ]

Single-site solution (mark base station and sketch coverage):



Two-site solution with relay (mark both sites and sketch coverage):



Clearly indicate site locations and approximate coverage regions.